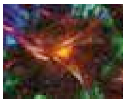


First quantum brain sensor built
Researchers build first modular quantum brain sensor, used it to record a brain signal. **https://dgit.in/jul21-16**

Self healing DNA
Astronauts watched as the DNA - cut with gene editing tools - repaired itself on the ISS. **https://dgit.in/jul21-17**

amine molecules, amino acids, and neuropeptides. So far, in the ongoing study of these chemical messengers, scientists have been able to discover a dozen small-molecule neurotransmitters and more than 100 different neuropeptides. Each different combination of these caters to a different function in the human body.

Whenever there needs to be communication established between two neurons, they generate an electric event, which is called an action potential. And, whenever a neuron gets excited due to the release of an impulse by the brain and creates this action potential, a particular type of neurotransmitter specific to the function of the associated neuron is released. There are hundreds of synapses at the end of a neuron, which are literally the points that allow a neuron to pass on an electrical signal from itself to the adjoined neuron. Whenever there is an action potential developed, each of the hundreds of synapses that are there at the end of the neuron release a neurotransmitter, which is the intermediary carrier of the electrical signal.

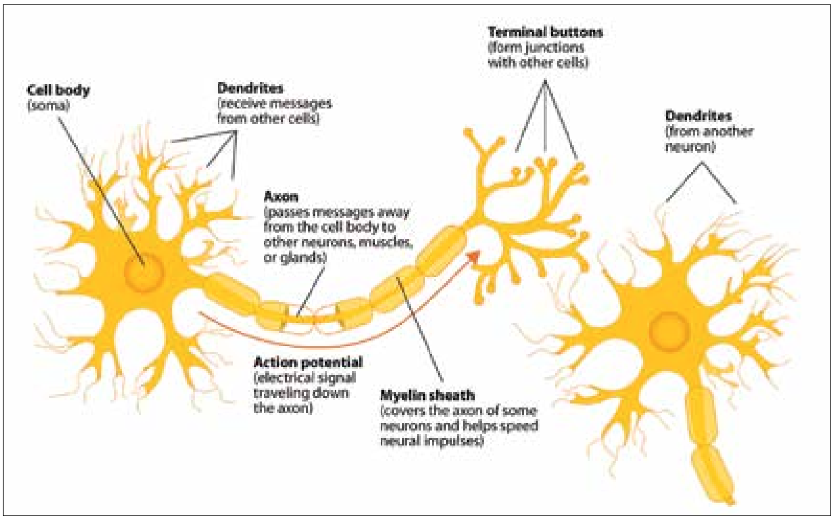
During this transmission of the impulse, a neurotransmitter can have one of three effects on a neuron. They are named excitatory, inhibitory or modulatory effects. Each of these have different functions, as we have mentioned earlier. An excitatory neurotransmitter promotes the creation of the action potential in the neuron which it is targeting, and the neurotransmitter acting in the opposite way to this is the inhibitory neurotransmitter. What they do is prevent or discourage the formation of an action potential on the receiving neuron. The last one of this group of three is the modulatory neurotransmitter. Their function is relatively complex to understand. Since they are not bound to any of the synaptic clefts of the neurons that they are generated in, they can affect a large number of neurons at once. They can distribute messages to several neutrons at once and affect other neurotransmitters' behaviour. One more important characteristic of these neurotransmitters is that they act at a relatively slower pace than their excitatory and inhibitory counterparts.

Now, the question that arises is what is the effect of these different types of neurotransmitters that can be experienced or felt by a normal human being. So let us start by throwing some names at you, starting with acetylcholine, dopamine, endorphins, and serotonin. Sound familiar, right? You may have read and heard about each one of these several times so far and may be familiar with how each of these affects our body. But did you know that each one of these is a type of neurotransmitter? Let's delve into the functions of these neurotransmitters.

Starting with one of the most talked about neurotransmitter, dopamine. This neurotransmitter plays a role in our perception of pleasure, while also playing a significant role in our ability to think and plan things and activities. There's a lot more to be read and understood about dopamine, but for now that much is enough. Moving on, we have serotonin. This neurotransmitter is responsible for controlling the human body's circadian rhythm, sleep, mood, and blood clotting. A disturbance in the level of serotonin in body is what causes the development of conditions such as depression and anxiety.

The next entrant is GABA. No, not the ground where India defeated Australia in Tests, but the thing that is present in your body. One of the firsts and only ones included here, that is an inhibitory neurotransmitter, GABA is responsible for mood regulations and stopping the neurons from getting over excited. Epinephrine is better known as adrenaline and is literally one of a kind. Responsible for controlling the body's fight or flight response, what makes adrenaline/endorphin unique is the fact that it is both a hormone and neurotransmitter. And, the last one that we will be explaining to you here is the endorphin class of neurotransmitters, that is the one that helps you recover from the pain that you experience when you jump off a ledge due to an adrenaline rush. The endorphins in our body are responsible for inhibiting pain signals, leading the body into a relatively energised and euphoric state.

The functions of human body are complex and the coordinators of it, the neurotransmitters aren't particularly easy to understand either. But, this was a small effort to give you the idea of how they work so that you can geek out the next time your friend says that he or she is feeling bummed out. d



Structure of a neuron